

Prime-ary Walsh splitting and proper cells in even dimension

Companion to walsh_positive_splitting.tex

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Status. Independently mathematically reviewed; not formally verified. These are continuous-density constructions. Neither theorem claims an improved finite-dice size bound. The proper-cell variant was independently proposed by two collaborating agents. The prime-ary variant increases the ratio of the total uniform drift to the drift spent in fully active cells.

Fix a prime q , a primitive q th root ζ , and a total-degree cutoff n . Use formal generators H and $Y_{i,a}$, where $i \in [n]$ and $a \in \mathbb{F}_q^*$, over $\mathbb{Q}(\zeta)$. Quotient the associative algebra by all words containing two Y letters with the same first index i , and by all terms of degree above n . For $b \in \mathbb{F}_q^n$, let $\rho_b(H) = H$ and $\rho_b(Y_{i,a}) = \zeta^{ab_i} Y_{i,a}$. Starting with $W_0 = \exp(H + \sum_{i,a} Y_{i,a})$, set

$$W_\ell = \prod_{t=0}^{q-1} \rho_{tb_\ell}(W_{\ell-1}),$$

in the displayed order. A nonzero monomial determines a frequency vector $v \in \mathbb{F}_q^n$: its coordinate is a if $Y_{i,a}$ occurs, and zero otherwise.

Lemma 1. *If B_ℓ has rows $b_1, \dots, b_\ell$, every log term with frequency $v \neq 0$ has degree at least $1 + \text{wt}(B_\ell v)$. The zero-frequency terms equal $q^\ell H$.*

Proof. At level zero, the assertion is immediate. In the BCH expansion for q factors, the linear sum multiplies frequency v by $\sum_{t=0}^{q-1} \zeta^{tb_\ell \cdot v}$, which is zero when the new coordinate is nonzero. A nonlinear term is a bracket of $k \geq 2$ components with disjoint first-index supports (otherwise it vanishes). Its degree is at least k plus the sum of the previous Hamming weights, which is at least $k + \text{wt}(B_{\ell-1}v)$. This allows the new required weight to increase by one. Zero frequency can occur for a nonzero monomial only when no Y occurs, since each first index appears at most once; all brackets composed only of H vanish. $\square$

Theorem 2. *For each fixed prime q , there are $q^{O_q(n)}$ equal rational cells supporting a permutation-fair family of continuous densities $f_i \in \{0, q\}$. The activity indicators on a uniformly selected cell are independent with probability $1/q$. In particular, the total mass contributed to each die by cells on which all n dice are active is exactly q^{1-n} .*

Proof. Choose a matrix $B \in \mathbb{F}_q^{L \times n}$ of rank n such that all nonzero images have weight at least n , with $L = O_q(n)$. For completeness, independent uniform rows give a binomial weight with mean $L(1 - 1/q)$ for every nonzero input. Taking L a sufficiently large constant multiple of n , a Chernoff bound and the union bound over $q^n - 1$ inputs prove existence. Lemma 1 gives $\log W_L = q^L H$. Substitute $H = \sum_i X_i$ and $Y_{i,a} = X_i$ in the reduced algebra. The substitution respects the repeated-first-index ideal. A factor indexed by $t \in \mathbb{F}_q^L$ has coefficient

$$1 + \sum_{a \neq 0} \zeta^{a(B^T t)_i} = q \mathbf{1}_{(B^T t)_i=0}$$

on X_i . Thus all velocities are real, nonnegative, and rational. After rescaling cell duration by q^{-L} , the signature is $\exp(\sum_i X_i)$. Surjectivity of B^T makes its values uniform on $\mathbb{F}_q^n$, which proves the activity and mass assertions. $\square$

Theorem 3. *For each even $n \geq 2$ and each fixed prime q , there is an exact fair continuous family with $q^{O_q(n)}$ equal rational cells, with densities in $\{0, q\}$, such that every cell has at least one inactive die.*

Proof. Choose the rows of B in the hyperplane $\mathbf{1}^\perp$, with

$$\ker B = \mathbb{F}_q \mathbf{1}, \quad \text{wt}(Bv) \geq n \quad (v \notin \ker B).$$

The same random-row argument on the quotient gives $L = O_q(n)$. The proof of Lemma 1 applies without injectivity: it kills every nonkernel frequency. A nonzero kernel frequency has every coordinate nonzero, so any surviving term uses all n distinct first indices and has degree exactly n , with no H letters. Hence

$$\log W_L = q^L H + E_n,$$

where E_n is homogeneous of degree n and is central after truncation. Apply a fixed phase $c \in \mathbb{F}_q^n$ with $\sum_i c_i \neq 0$ before substitution. The factors now have activity characterized by $(B^T t + c)_i = 0$. Since every $B^T t$ has coordinate sum zero, no factor has all coordinates active. The log still has the form $q^L H + E'_n$.

Append the same list of exponential factors in reverse order. Reversing a product of exponentials of degree-one elements multiplies its degree- d log term by $(-1)^{d+1}$: word reversal is an antiautomorphism and acts by that sign on a homogeneous Lie polynomial. Since n is even, the reversed log is $q^L H - E'_n$. Centrality now shows that the concatenation has log $2q^L H$ exactly. Normalize the cell durations by $1/(2q^L)$. All cells remain proper. $\square$

Corollary 4. *For $q = 2$ and even n , the final reversal in Theorem 3 can be omitted by choosing the first row of B to be $\mathbf{1}$.*

Proof. That row lies in $\mathbf{1}^\perp$ because n is even. Add enough independent random rows afterwards to obtain the same distance property. Setting $H = 0$ in the first symmetrization gives $\exp(\sum_i Y_i) \exp(-\sum_i Y_i) = 1$. Every later symmetrization, and every fixed phase, preserves this identity. Thus every nonzero log error contains at least one H . A surviving nonzero kernel frequency also requires all n distinct Y_i , so its degree exceeds the cutoff n . $\square$

Limits. Theorem 3 allows activity as large as $n - 1$, so naive recursive finiteization still gives a poor recurrence. Theorem 2 supplies a drift ratio q^{n-1} , but a quantitative positive absorption lemma is still required to remove its full cells. The cell count measures geometric complexity of continuous densities; it is not the face count of a finite equiprobable die.